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Methodology

Toward a Smaller Design for EQ-5D-5L Valuation Studies

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ABSTRACT

Background: To construct an EQ-5D-5L value set, the EuroQol Group developed a standard protocol named EuroQol Valuation Technology (EQ-VT), prescribing the valuation of 86 health states utilizing the composite time trade-off (cTTO) approach, and subsequently modeled the observed values to yield values for all 3125 states.

Objective: A recent study demonstrated that a 25-state orthogonal design could provide as accurate predictions as the EQ-VT design applying visual analogue scale data. We aimed to test that design using time trade-off (TTO) data.

Method: We collected TTO values utilizing EQ-VT, orthogonal, and D-efficient designs. The EQ-VT design included 86 health states distributed over 3 blocks of 30 states with some duplicates. The orthogonal and D-efficient designs each comprised 1 block of 30 states. A total of 525 university students were asked to value a random block of health states using EQ-PVT (a PowerPoint replica of EQ-VT software), which generated 100 observations per health state in all 3 designs. We modeled data by design and compared the root mean square error (RMSE) between observed and predicted values within and across the designs.

Results: The EQ-VT design had the lowest RMSE of 0.052; the RMSEs for the orthogonal and the D-efficient designs were 0.066 and 0.063, respectively. RMSE results between designs differed for more severe health states. Some coefficients differed between designs.

Conclusion: Smaller designs did not lead to significant increases in prediction errors when modeling TTO data (measuring 0.01 on a utility scale). Resource-constrained countries may use small designs for valuation studies, especially when other types of preference data, such as those from discrete choice experiments, are collected and modeled jointly.

Keywords: EQ-5D-5L, small design, TTO, valuation.

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Introduction

To estimate an EQ-5D-5L value set, the EuroQol Group developed the standardized EuroQol Valuation Technology (EQ-VT) protocol.[1] The protocol design includes the selection of 86 different EQ-5D-5L health states. These 86 states are arranged into 10 blocks of 10 health states for the composite time trade-off (cTTO) task. This task includes 2 parts: (1) using the conventional time trade-off (TTO) task to derive values above 0, and (2) using a 10-year lead-time TTO task to derive values below 0.[2] To achieve adequate precision for the mean values, it was decided that each block should have 100 observations. The optimal sample size was estimated to be around 1000 respondents (10 blocks × 100 observations per block).[3,4] With this design, EQ-5D-5L value sets for 15 countries or regions have been established and published.[5-19]

The selection of 86 states for direct valuation through cTTO rested on considerations of robustness, requiring that each parameter be derived from multiple stimuli. Furthermore, this approach would enable some interaction terms to be included in the regression model while still allowing for 1 degree of freedom per parameter. However, the latter consideration no longer carries much weight because practical experience and the results from a number of simulations have suggested that the main effects model (where no interactions between health dimensions were considered) with 21 parameters (5 × 4 dummy variables + intercept) performs well,[5,20] leaving a surplus of 65 degrees of freedom. The reduced need for a large design has raised the question of whether a small design can be used to promote the feasibility of a valuation study.

Assuming the number of observations per health state is fixed (eg, 100 in the case of the EQ-VT protocol), then the sample size required in a valuation study depends on the size of the design—that is, the number of health states that need to be directly valued. The benefit of adopting a small design is that the sample size

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needed for a valuation study can be minimized. Notably, if the purpose of a value set is to guide decision making, a criterion in adopting the small design would be to not compromise sample representativeness.

Apart from representativeness, another important technical criterion in adopting a small design is that the modeling results of a small design should not compromise prediction accuracy at an unacceptable level. Two recent studies have addressed the impact of design size and the approach to design generation on the accuracy of predicted values using saturated EQ-5D-5L visual analogue scale (VAS) valuation data sets.[21,22] A saturated data set contains observed VAS values for all EQ-5D-5L states, allowing predictive errors associated with different designs to be quantified. The research strategy taken in these studies was that, initially, different candidate designs were identified (eg, the EQ-VT design or an orthogonal design). Then, the health state values of each candidate design were modeled to predict the values of all EQ-5D-5L states. Finally, the prediction errors of different design choices were quantified (computed as the root mean square errors [RMSEs] of the predicted values against the observed values) and compared. The results suggested that the current EQ-VT design performed well among different design choices. However, it is worth noting that the orthogonal design with only 25 health states performed as well as the EQ-VT design, assuming the main effects model to be sufficient. One shortcoming of the orthogonal design was that it had larger mispredictions for the mild states (ie, states with higher VAS values) compared to the EQ-VT design.[12] Another study comparing design choices for the 3-level EQ-5D VAS saturated data set also indicated the superiority of the orthogonal design.[23]

A limitation of the previously-mentioned studies is that VAS was used to collect observed values. Generalizability of the results to valuation data collected using the cTTO method is an open question. VAS values have a different distribution compared with cTTO values. For example, VAS values do not display the relatively linear heteroscedasticity typically observed in TTO values. In addition, a property of VAS is “end of scale aversion.”[24,25] Although the maximum attainable score is 100, many people hesitate to assign such a high score to any state. It is unclear whether this end of scale aversion phenomenon partially accounts for the large mispredictions around mild states observed in VAS data. In the present study, we aimed to investigate whether a small design can be used for EQ-5D-5L valuation using cTTO without increasing mispredictions in any part of the severity scale to unacceptable levels.

Methods

Strategy

We collected TTO data for 3 designs with, in total, 500 students. The 3 designs (the standard EQ-VT design and 2 candidate small designs) were divided into 5 blocks and distributed across the respondents, ensuring a minimum of 100 observations per block. Each student valued 1 block of 30 EQ-5D-5L health states using the cTTO method. We modeled the observed TTO data for each design to predict values for all possible EQ-5D-5L health states. The predictive accuracy of the data derived for each design was compared with the observed values by calculating the RMSE.

Experimental Designs

The study comprised 3 designs that differed in terms of health states included (see the Appendix in the Supplementary Materials found at https://doi.org/10.1016/j.jval.2019.06.008):

- the EQ-VT design including 86 health states.
- an orthogonal design including 25 health states.
- a D-efficient design including 25 health states.

The design of EQ-5D-5L valuation studies has been standardized across countries. It includes cTTO tasks for 86 EQ-5D-5L health states. We refer to this set as the EQ-VT design. Oppe et al.[3] provided a detailed description of the TTO task in EQ-5D-5L valuation studies. Briefly, a data set of 1000 respondents with values of the 3125 possible EQ-5D-5L states was created via simulation, using a set of priors based on the findings of a multinational pilot study. Next, 10,000 sets of 86 health states were created, taking level balance into account and ensuring the worst EQ-5D-5L state and the 5 mildest states were always included. Ordinary least squares regressions were performed on these sets of 86 health states, and the resulting parameter estimates were compared with the priors. The design with the lowest mean square error was kept as the final design.[3]

An orthogonal design was generated by assigning dimensions and levels defined by EQ-5D-5L to a preexisting orthogonal array. For EQ-5D-5L, an orthogonal main-effect design mathematically contains a minimum of 25 health states. A theoretical advantage of orthogonal designs is that the absence of correlations between dimensions offers a strong basis for decomposing the observed values to underlying dimension severity levels under the assumption that the model is specified correctly. Optimal orthogonal designs are also level balanced. That is, each level occurs equally often within each dimension. In this study, we used the best-performing orthogonal design from the previous VAS saturated study, in which 100 variants of orthogonal designs were created and tested with respect to their prediction performances for all 3125 EQ-5D-5L health states.[21] To account for the previously mentioned issue concerning the prediction of values for mild states in orthogonal designs, the 5 mild states were added to the orthogonal array. The design's performance was investigated with and without the 5 mild states.

A D-efficient design aims at minimizing the D-error, which is the determinant of the inverse of the Fisher information matrix.[26] Unlike the orthogonal design with a fixed number of health states, the minimum number of health states in the D-efficient design depends on the required degrees of freedom. For comparison reasons, the number of unique states sampled for the design was set at 25, and the 5 mild states were added for a total of 30 states. Furthermore, the design can be generated subject to constraints (eg, to avoid implausible combinations between dimensions) or to include the mildest states (4 level 1, 1 level 2) in the design. Hence, the D-efficient design is more flexible if constraints are considered. In developing the D-efficient design, we generated 40,000 different designs with reasonable level balance (level balance criterion $\leq$ 18)[4] and kept the best 400 candidates in terms of D-error and level balance. Next, we validated these best 400 candidates in a saturated EQ-5D-5L VAS data set[27] and a simulated saturated data set based on the Chinese EQ-5D-5L TTO value set.[6] The design with the lowest number of prediction errors was selected for this study.

Blocking

The set of 136 (86 + 25 + 25 states) health states was divided into 5 blocks. The EQ-VT design was divided into 3 blocks. To arrive at the same block size of 30, some health states were duplicated across blocks, with each block containing the 55555 state and at least 2 of the mildest states. The orthogonal design and the D-efficient design each formed 1 independent block of 30 states. Each respondent valued 1 block.

Interviewers and Respondents

Following sample size consideration explained by Oppe et al,[1] we collected TTO values for the 5 blocks of health states from 500 university students at Guizhou Medical University, China. We used students as respondents because the large number of health states to be valued (30 states per block) would be overdemanding for a general population sample.

Data were collected by 7 interviewers facilitated by 1 respondent coordinator. All of the interviewers and the respondent coordinator were senior students from Guizhou Medical University. Before data collection, all interviewers and the coordinator received 3 days of training with respect to quality-adjusted life years, the EQ-5D questionnaire, and the EQ-PVT software to help them understand the context of the TTO questions. For the interview location, we rented 4 offices from Guizhou Medical University and set up 2 interview stations in each office. Each interviewer was assigned to a fixed interview station, and all of his or her interviews were conducted at that station. The recruitment advertisement was circulated to university students through e-mails so that interested respondents could contact the coordinator. The coordinator then arranged appointments for the interviewers and respondents. Each respondent was paid 100 RMB (equivalent to 14 euros) upon successful task completion.

All interviews were conducted using EQ-PVT, which is a PowerPoint replica of the EQ-VT software and was obtained from the EuroQol Research Foundation. The EQ-PVT used in this study was consistent with the EQ-VT protocol version 1.1.[28] There were 3 steps in each interview. First, the background of the study was introduced, and the respondents were asked to give written consent to take part. Second, the respondents provided personal background information and reported their health states using EQ-5D-5L. Third, 36 health states consisting of 3 wheelchair examples, 3 practice EQ-5D states, and a randomly selected block of 30 EQ-5D states were valued through face-to-face interviews. Thus, before valuing the block of 30 EQ-5D-states, following EQ-VT protocol V1.1,[28] each respondent was familiarized with the TTO task using 3 wheelchair example states (wheelchair, a situation worse than a wheelchair, a situation better than a wheelchair) and 3 examples of EQ-5D-5L states (21121, 35554, 15411).[28] The 3 wheelchair examples were used to familiarize respondents with the cTTO approach in general and also to introduce them to the procedure by which they would value a health state as worse than death. The 3 practice states were selected to represent different severity levels of EQ-5D-5L states (21121 = mild, 35554 = severe, 15411 = moderate). No feedback or debriefing module was provided to the respondents.

Interviews were conducted in accordance with current EuroQol guidelines for quality control as reported by Ramos Goni et al.[29] Quality control reports indicating protocol compliance and the presence of interviewer effects were sent to the interviewers every 3 days with individual suggestions on how to improve interview performance, if necessary.

Data Analysis

Because we fixed the number of observations for each health state at 100, the EQ-VT design had 3 blocks $\times$ 30 states $\times$ 100 observations, or 9000 observations in total, and the 2 small designs each had 30 states $\times$ 100 observations, or 3000 observations in total. The data were analyzed by design using 2 different model specifications plus 2 different standard error specifications, based on the original set (25 states) or the extended set (30 states). To resolve the large misprediction issue found in the VAS study, we explored 2 approaches: (1) to extend the original set with the 5 mildest states, and (2) to test the assumptions on how standard errors were specified in the model (heteroscedasticity vs homoscedasticity). For model specifications, we compared the performance of a 20-parameter additive model (Equation 1) and an 8-parameter multiplicative model (Equation 2). For the 20-parameter additive model, cTTO utility was explained by 20 dummy variables and 1 intercept. For each dimension (MO for mobility, SC for self-care, UA for usual activity, PD for pain/discomfort, AD for anxiety/depression), 4 dummy variables were used to represent the disutility from level 1 to the other 4 levels (eg, MO_3 took 1 if the health state had a problem in the third level of mobility, and it took 0 if otherwise).[30]

$$\begin{aligned}\text{Utility} = {} & \alpha+\beta_1 MO_2+\beta_2 MO_3+\beta_3 MO_4+\beta_4 MO_5+\beta_5 SC_2+\beta_6 SC_3 \\ & +\beta_7 SC_4+\beta_8 SC_5+\beta_9 UA_2+\beta_{10} UA_3+\beta_{11} UA_4+\beta_{12} UA_5 \\ & +\beta_{13} PD_2+\beta_{14} PD_3+\beta_{15} PD_4+\beta_{16} PD_5+\beta_{17} AD_2+\beta_{18} AD_3 \\ & +\beta_{19} AD_4+\beta_{20} AD_5+\varepsilon\end{aligned} \tag{1}$$

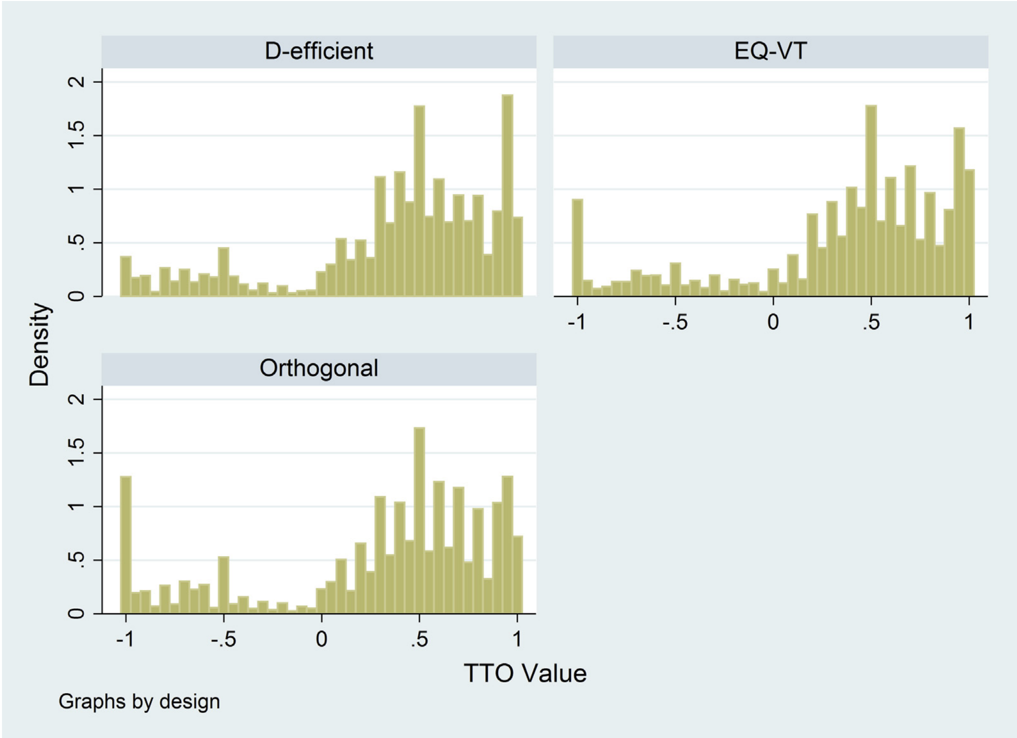
The 8-parameter multiplicative model is a variant of the 20-parameter model. It rests on the assumption that the relative distance between levels is the same across all dimensions.[31] In this model, 5 dimension parameters were used to represent the disutility of having problems at level 5 on each dimension (β_{MO}, β_{SC}, β_{UA}, β_{PD}, and β_{AD}); a set of scalars L2-L4 (ie, level parameters) was estimated to identify where the cutoffs of the intermediate levels were located, subject to the constraint that the relative distance between levels was constant across all dimensions. The most severe level, L5, was set to 1 and not estimated in the model. Thus, the absolute amounts of the utility of the dimension severity levels were computed by multiplying the relevant scalar L2-L4 with the dimension weights. For example, the disutility of level 4 on anxiety was β_{AD} x L_4 and the disutility of level 5 on mobility was β_{MO}.

$$\begin{aligned}\text{Utility} = {} & \alpha+(\beta_{MO}x_{MO2}+\beta_{SC}x_{SC2}+\beta_{UA}x_{UA2}+\beta_{PD}x_{PD2}+\beta_{AD}x_{AD2})L_2 \\ & +(\beta_{MO}x_{MO3}+\beta_{SC}x_{SC3}+\beta_{UA}x_{UA3}+\beta_{PD}x_{PD3}+\beta_{AD}x_{AD3})L_3 \\ & +(\beta_{MO}x_{MO4}+\beta_{SC}x_{SC4}+\beta_{UA}x_{UA4}+\beta_{PD}x_{PD4}+\beta_{AD}x_{AD4})L_4 \\ & +(\beta_{MO}x_{MO5}+\beta_{SC}x_{SC5}+\beta_{UA}x_{UA5}+\beta_{PD}x_{PD5}+\beta_{AD}x_{AD5})L_5+\varepsilon\end{aligned} \tag{2}$$

Rand-Hendriksen et al[31] reported no difference in performance in terms of out-of-sample predictions between the 8- and 20-parameter models on Spanish, Singaporean, and Chinese EQ-5D-5L valuation data. The 8-parameter model required fewer degrees of freedom compared with the 20-parameter model and perhaps a better model specification for small designs. To address the question with respect to misprediction of the values for mild states, we ran these models by using 3 designs with and without the 5 mildest states and under the assumption of either homoscedasticity (random effect generalized least squares) or heteroscedasticity (heteroscedastic regression).

To judge design performance, we (1) predicted health state values by modeling the 3 designs respectively and computed the RMSE within the design and the RMSE across designs; (2) compared the similarity of coefficients between designs; and (3) compared the RMSE along the misery indexes to see whether the mispredictions depended on health state severity. The misery index is the sum of 5 digits of an EQ-5D health state (eg, the misery index of state 13255 is 1 + 3 + 2 + 5 + 5 = 16) and was

Figure 1. Time trade-off (TTO) values distribution across three designs.



used as a proxy for health states' severity levels. Considering that we had multiple models/designs, in this article we report the RMSE results first and then use only the best model for the subsequent analysis. Additionally, we provide the observed values and predicted values of 10 states for reference. Designs were created using STATA 14 (StataCorp LLC, College Station, TX) and R 3.4.2. Analyses were performed using STATA 15 (StataCorp LLC).

Results

Raw Data

A total of 557 interviews were completed. For quality reasons, the first 32 interviews of 1 interviewer were dropped. After retraining, this interviewer reconducted 32 interviews with new respondents. Data for the unqualified 32 interviews were not analyzed. On average, each interview took about 46 minutes (SD 14 mins). Each respondent made around 7.89 moves per state (SD 2.11 moves) on the valuation task, including the 6 practice states. The average time to complete a single TTO task was 56.9 seconds for the non-practice states. The observed values for all health states can be found in the Appendix (see Supplementary Materials found at https://doi.org/10.1016/j.jval.2019.06.008). The highest mean value was 0.950 for state 12111, and the lowest was -0.719 for state 55555. More severe states had larger standard deviations.

Figure 1 shows similar data distributions for the EQ-VT design and the orthogonal design, whereas the D-efficient design had fewer observations clustering at -1 but more observations clustering at 0.95. In general, the values were distributed over the whole valuation space, with more positive values provided. Digital preferences were shown as more clustering at integral numbers such as 1 and 0.5.

Modeling and Prediction Performance

Table 1 indicates that the EQ-VT design performed better than the 2 small designs in terms of the overall RMSE (0.053). The choice of model specification and standard error specification did not have much impact on the RMSE results. Notably, extending to include the 5 mildest states lowered the overall RMSE for the D-efficient designs (from 0.095 to 0.063) but not for the orthogonal design (0.066 vs 0.067). Next, we report the results of the 3 designs using the 8-parameter multiplicative model extended to include the 5 mildest states for the 2 small designs. Table 2 shows the regression output of the 8-parameter model. In general, the D-efficient design had the largest level parameters (L2, L3, and L4), and the orthogonal design had the smallest level parameters. The 95% confidence intervals did not overlap for the dimension parameters of mobility (EQ-VT vs orthogonal) and anxiety (D-efficient vs orthogonal and EQ-VT) or for the level parameter of level 4 (orthogonal vs EQ-VT and D-efficient). In terms of the RMSE, small designs had better predictions for the empirical state set than the EQ-VT design. EQ-VT had the lowest RMSE for all states (0.053), and the RMSEs of the 2 small designs did not differ much (0.066 vs 0.063).

Figure 2 shows that all 3 designs performed similarly in mild states (ie, with a maximum misery index of 13). The EQ-VT design performed evenly along the scale except for group 23. In that group, only 1 state (55535) that belonged to the orthogonal design occurred in our data set. The predicted value for 55535 using the EQ-VT design was -0.343, whereas the actual value observed was -0.617. The orthogonal design performed worse in the 2 misery index groups 14 and 17, whereas the D-efficient design did not perform well in the severe part of the scale (misery index >20). Table 3 shows the predicted and observed values of 10 random health states.

Table 1. RMSE full results.

Model	All	EQ-VT	Orthogonal + 5	Orthogonal	D-efficient + 5	D-efficient
No. of health states	146	86	30	25	30	25
Multiplicative	0.051	0.053	0.066	0.067	0.063	0.067
GLS additive	0.049	0.053	0.069	0.069	0.063	0.095
Hetero additive	0.051	0.054	0.064	0.072	0.065	0.092

EQ-VT indicates EuroQol Valuation Technology; GLS, generalized least squares; hetero, heteroscedastic regression; RMSE, root mean square error.

Discussion

This study built on a previous EQ-5D-5L VAS valuation study that found that statistically efficient small designs could be used for valuation studies without compromising prediction accuracy.[21] As in the VAS study, our study using the TTO approach showed that the EQ-VT design performed best in terms of prediction accuracy. Smaller designs had higher RMSEs, but the differences were relatively small, measuring 0.01 on a utility scale. It should be noted that in the present study the comparisons were made for only 136 states instead of all 3125 states covered in the EQ-VAS study. Because the EQ-VT design accounted for 63% (86/136) of all states, the overall prediction result was more advantageous toward EQ-VT. In addition, using the same model, the parameters of the 3 designs differed considerably. There are 2 possible reasons for this: (1) a different design (subset of health states) might have produced a different set of coefficients; (2) respondents may have differed in health preferences across the 3 designs. Further study is needed to understand this issue.

It is notable that we did not encounter the large misprediction problem found in the EQ-VAS study, where the RMSE of the mild states (VAS >70) was twice the size as that for the other states.[21] A possible explanation for this misprediction in the VAS study was the large gap between the value of 11111 and the values of any other states.[21] The relative magnitude of the RMSE in the present study was similar to that reported in the VAS study. In the latter, the RMSEs for all 3125 states were 3.44 and 3.87 (about 3.44/100, or 3.4%, on a VAS scale) for the EQ-VT and orthogonal designs respectively, and the counterparts were 0.053 and 0.066 (about 0.053 of 2, or 2.7%, on a utility scale) in the current study.

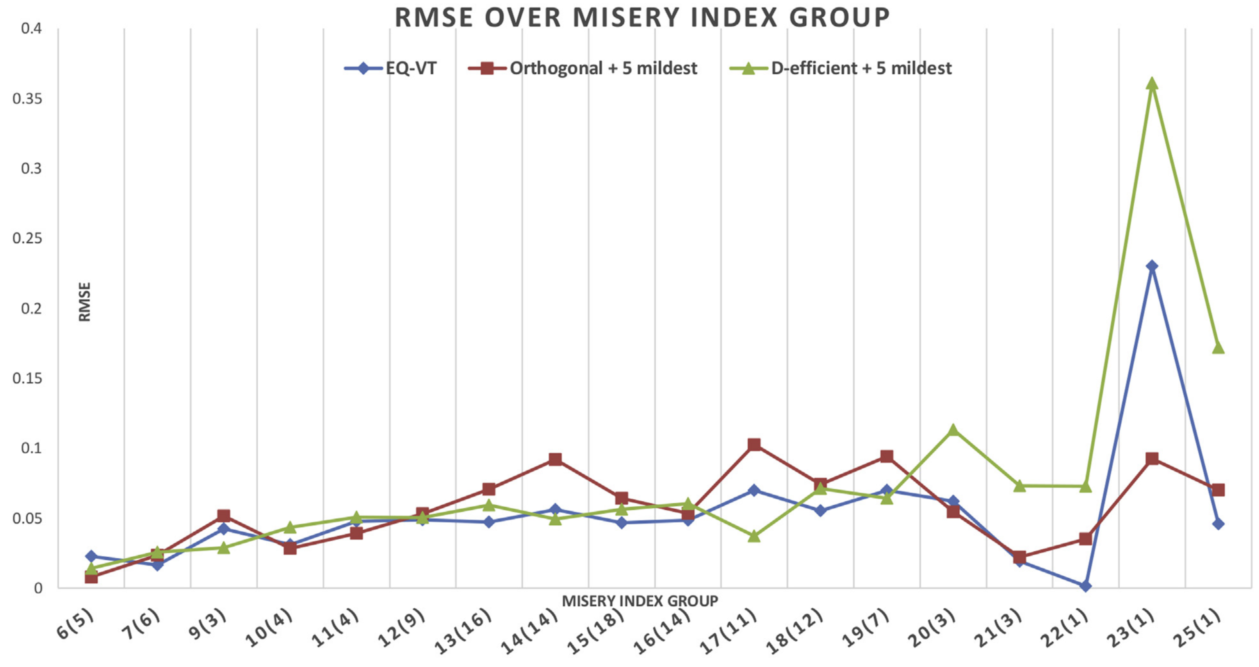
With respect to the 20-parameter model, the coefficients of all 5 level-2 dummy variables were not significant for the 2 small designs. One possible explanation is that the 2 small designs had insufficient power owing to small sample sizes. Increasing the degrees of freedom by extending the small designs with the 5 mildest states and using an 8-parameter model improved this issue for the D-efficient design. This explained the improved prediction performance of the D-efficient design when using the 8-parameter model extended to include the 5 mildest states. However, these 2 approaches did not work on the orthogonal design. The different effects of the mild states extension may be explained by the different basis on which these 2 designs were constructed. In the orthogonal design, the extension disrupted orthogonality and level balance. By comparison, the focus of the D-efficient design was on standard errors around parameters. Hence, if the number of states were extended, modeling results could improve because extra states could provide information on the parameters difficult to estimate. Future research could increase the sample size for small designs to provide further understanding of this issue.

Table 2. Multiplicative regression model output and RMSE results.

Model coefficients	Bayesian + 5 mildest states	Orthogonal + 5 mildest	EQ-VT
	Coe (95% CI)	Coe (95% CI)	Coe (95% CI)
Constant	0.961 (0.920, 1.002)	0.943 (0.904, 0.983)	0.992 (0.961, 1.023)
Mobility	−0.280 (−0.335, −0.225)	−0.363 (−0.402, −0.325)	−0.296 (−0.320, −0.272)
Self-care	−0.207 (−0.254, −0.161)	−0.270 (−0.309, −0.232)	−0.244 (−0.269, −0.219)
Usual activities	−0.275 (−0.314, −0.235)	−0.330 (−0.368, −0.291)	−0.294 (−0.319, −0.268)
Pain/discomfort	−0.425 (−0.464, −0.385)	−0.356 (−0.395, −0.317)	−0.404 (−0.430, −0.379)
Anxiety/depression	−0.322 (−0.359, −0.284)	−0.413 (−0.452, −0.374)	−0.428 (−0.453, −0.402)
Level 2	0.127 (0.065, 0.188)	0.036* (−0.017, 0.089)	0.115 (0.077, 0.152)
Level 3	0.316 (0.261, 0.370)	0.258 (0.211, 0.304)	0.293 (0.259, 0.326)
Level 4	0.845 (0.778, 0.913)	0.674 (0.621, 0.726)	0.780 (0.748, 0.812)
Design Characteristics and performance			
No. of health states	30	30	86
RMSE for empirical state set	0.024	0.036	0.051
RMSE for validation state set	0.070	0.072	0.054
RMSE for all states	0.063	0.066	0.053

CI, confidence interval; EQ-VT indicates EuroQol Valuation Technology; RMSE, root mean square error.
*Not significant on .05 level.

Figure 2. Root mean square error (RMSE) of three designs over severity scale.



Furthermore, we fixed the number of observations per health state at 100 and sampled a small subset of health states. Another approach would be to sample more health states with fewer observations per state. Although our approach could provide more robust observed mean values for directly valued health states, the other approach could cover a wider valuation space. Further research is needed to understand the differences in performance of these approaches in estimating value sets.

An open question is whether using different designs would result in different results in cost-utility analysis. In previous research, the use of different EQ-5D value sets led to different incremental cost-effectiveness ratios.[32] In our study, the coefficients for each design differed in the value sets generated, thus estimating different predicted health state utilities. Future research could evaluate the effect of adopting small designs in the context of cost-utility analysis.

With an increasing number of national health technology assessment agencies preferring national value sets,[33] more valuation studies of various instruments will be conducted. How to sample health states is a key design issue in valuation studies for all preference-based multi-attribute instruments such as the Short Form of Six Dimensions (SF-6D),[34] the Asthma Quality of Life Questionnaire (AQLQ),[35] the QualiPause Inventory (QPI),[36] and the European Organization for the Research and Treatment of Cancer Quality of Life Questionnaire (EORTC QLQ-C30).[37] Our study shows that the use of small designs improves the feasibility of a valuation study while not compromising its validity. This offers a more efficient approach to undertake valuation studies for countries with lower budgets, for example, and for methodological research. Nevertheless, we do not immediately expect the deployment of small designs as the new default choice, mainly because using a large and representative sample is also important in a valuation

Table 3. Observed values and predicted values for 10 random health states.

Health state	Observed value	Standard error	Lower 95% CI	Upper 95% CI	Predicted by EQ-VT	Predicted by Orthogonal + 5	Predicted by D-efficient + 5
11221	0.915	0.007	0.901	0.930	0.912	0.916	0.847
12324	0.502	0.038	0.427	0.578	0.498	0.557	0.518
24422	0.437	0.035	0.368	0.505	0.443	0.498	0.421
31514	0.362	0.040	0.283	0.441	0.278	0.241	0.324
34234	0.311	0.041	0.230	0.392	0.229	0.286	0.263
43154	−0.017	0.050	−0.115	0.082	−0.048	−0.005	−0.034
43542	−0.009	0.055	−0.117	0.010	0.032	0.045	−0.006
44355	−0.365	0.048	−0.461	−0.269	−0.347	−0.336	−0.271
55424	−0.259	0.056	−0.370	−0.147	−0.157	−0.203	−0.073
55535	−0.617	0.041	−0.699	−0.536	−0.387	−0.524	−0.239

CI indicates confidence interval; EQ-VT, EuroQol Valuation Technology.

study, especially when the result is used repeatedly in the decision-making process.

This study had limitations. First, the TTO task to be completed by each respondent included 30 EQ-5D-5L health states plus another 6 exercise states. The respondent burden was almost tripled compared with the standard EQ-VT protocol. Nevertheless, by applying the quality control procedure during data collection,[29] we did not observe any evident data clustering in our study like that shown in earlier EQ-5D-5L valuation studies.[8,12] Further, our respondents were university students who had fewer socioeconomic differences and whose health preferences were more homogeneous. Studies using the general public for sampling purposes could increase study sample size to achieve the same level of prediction accuracy.

The study findings offer support for the use of the current EQ-VT design. It is difficult to tell precisely what the results mean for smaller designs because the level of concern with the change in the RMSE essentially rests on an arbitrary evaluation. Because the relative increase of the RMSE was modest, we do not feel that the results raise a red flag over the use of small designs, especially if one considers that valuation researchers have other options to promote the robustness of their results. An established way to reduce the risk of errors in predicted values involves combining multiple types of valuation data. For instance, the EQ-VT protocol has a discrete-choice experiment (DCE) task included alongside the cTTO task, and a hybrid model can be used to model DCE data and cTTO data together.[38] The use of the hybrid model has proved superior to using the TTO data alone.[10-14,38] Such an approach offers a way to manage the risks of the larger errors in predicted values associated with smaller TTO designs while not undoing the benefits of shrinking the TTO design because collecting DCE responses is less resource demanding. This also paves the way to improve sample representativeness for a valuation study because the resources saved by using a small design could then be put toward, for example, using a probabilistic sampling method or collecting online DCE data to reach respondents from remote areas.

Conclusion

The EQ-VT design had the best prediction performance of the 3 designs studied and should be used as the default design for EQ-5D-5L valuation studies. Smaller designs also performed reasonably well and may be considered for use in some specific contexts such as for methodological research, in resource-constrained countries, and where data collection and modeling are paired with other valuation approaches such as DCE.

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Supplemental Materials

Supplementary data associated with this article can be found in the online version at https://doi.org/10.1016/j.jval.2019.06.008.

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